Question 4

GH-300: Security & Safety

How does GitHub Copilot mitigate the risk of accidentally outputting hardcoded secrets (API keys, passwords)?

A. By replacing hardcoded secrets with synthetic placeholder tokens.

✔ **B. By using built-in security filters that block suggestions containing recognized secret patterns.**

C. By encrypting secret strings prior to outputting them into code editors.

D. By automatically committing keys to GitHub Secrets.

Correct Answer: B

GitHub Copilot features automated AI safety and vulnerability filters. When generating code completions, the model scans output against patterns for credentials and API keys, automatically blocking suggestions containing identifiable secrets.

Option Breakdown

- **A. Replacing secrets with synthetic tokens:** Incorrect. While developers are encouraged to use placeholders, Copilot suppresses secret matches altogether rather than substituting them.
- **B. Built-in security filters:** Correct. Copilot uses real-time pattern matching filters to prevent credentials from being emitted in suggestions.
- **C. Encrypting secret strings:** Incorrect. Copilot never inserts encrypted secret payloads into editable source files.
- **D. Committing keys to GitHub Secrets:** Incorrect. Copilot does not interact with repository infrastructure or secrets management APIs.

💡 **GH-300 Exam Tip**
Copilot filters work on both inputs and outputs: pre-processing checks context while post-processing filters block security vulnerabilities and secret disclosure before suggestions appear on screen.